Course Name: Engineering Physics
Course Code: PH-101
Course Type: Institute Core

Contact Hours/Week: 3 **Course Credits:** 3

Course Objectives

- To create an ability to understand laser system, optical fibre in industries, laboratories and in communication
- To understand concepts of communication through electrodymanics.
- The broad education necessary to understand behavior of semiconductor devices.
- Acknowledge of concepts / technologies like superconductivity.

Unit Number	Course Content	Contact Hours
UNIT-01	**Electrostatics and Electrodynamics:** Gauss's Law in dielectric medium, Equation of continuity, displacement current, Maxwell's equations, wave equation for electromagnetic radiation, electromagnetic wave propagation in free space and isotropic dielectric medium, Poynting theorem & Poynting vector.	06L
UNIT-02	**Laser Physics:** Concepts of laser, spontaneous and stimulated emission, elementary idea about Lasers, basic principles involves in laser, three and four level laser system, coherence, characteristics of laser light; ruby, He-Ne, CO2 and semiconductor lasers, application of lasers.	06L
UNIT-03	**Fibers Optics and Photonics:** Optical Fiber, physical structure and basic theory, modes in optical fibers, step index and graded index fibers, losses in optical fibers, sources and sensors for optical fibers, applications of optical fibers in communication.	06L
UNIT-04	**Quantum Mechanics:** Need of quantum mechanics, Compton effect, Born's concept of wave function, Eigen function and Eigen values, operators in quantum mechanics, expectation values, time independent, time dependent Schrodinger's wave equations and its applications viz., particle in one dimensional potential well.	06L
UNIT-05	**Introduction to Solids:** Free electron theory (Classical & Quantum): Assumptions, Merits and drawbacks, Fermi level, Density of states, Periodic potential, Bloch's theorem, Kronig–Penny modal, E–K diagram, Effective mass, Origin of energy bands in solids, Classification of materials : Metals, semiconductors and insulators.	06L
UNIT-06	**Superconductivity:** Introduction and discovery of superconductivity, superconducting materials, Meissner effect, critical magnetic field and critical current, type-1 and type-2 superconductors, isotope effect, theory of superconductivity.	06L

Course Outcomes

Upon successful completion of the course, the students will be able to

CO1: Describe the optical devices and their applications.
CO2: Identify the applications of electrodynamics using Maxwell equations.
CO3: Apply concept of semiconductor physics to understand electronic systems.
CO4: Apply concepts of Quantum mechanics in solving physics problems at nanoscale.
CO5: Learn the working of equipment based on physical phenomenon.

Text Books:

1. M. N. Avadhanulu and P. G. Kshirsagar, A Textbook of Engineering Physics by S. Chand.
2. P. S. Aithal and H. J. Ravindra,Textbook of Engineering Physics by Acme Learning Pvt. Ltd., New Delhi.
3. Kittle C,State Physics by John Wiley & Sons, 2005.

References Books:

A. K. Ghatak and K. Thyagarajan, Lasers Fundamentals and Applications by Springer, 2010.
B. G. Streetman, Solid State Electronic Devices by Prentice Hall of India, New Delhi 2006.
A.S. Vasudeva, Modern Engineering Physics by S. Chand & Co. Ltd.
D.J. Griffiths,Introduction to Electrodynamics by Pearson Education Pvt. Ltd., New Delhi, 2002
A. Ghatak and S. Lokanathan,Quantum Mechanics by Mc Millan India Ltd.